Web Service Endpoint Verification

From an SSH session inside the Ubuntu VirtualBox VM, I sent requests to `/health` and `/` on port 8080. Both returned HTTP 200. The health response reported `ok`, and the root response included the service name, message, container hostname, and request count.

```
ubuntu@cs554-compose:~/cs454-midterm-project$ curl -i http://127.0.0.1:8080/
health
HTTP/1.1 200 OK
Content-Type: application/json
Date: Wed, 21 Oct 2026 03:00:08 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"status":"ok"}ubuntu@cs554-compose:~/cs454-micurl -i http://127.0.0.1:8080/
HTTP/1.1 200 OK
Content-Type: application/json
Date: Wed, 21 Oct 2026 03:00:21 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"service":"cs554-midterm","message":"Hello from my midterm service!","hostn
ame":"51a355a22d67","requestCount":1}ubuntu@ubuntu@cs554-compose:~/cs454-mid
ubuntu@cs554-compose:~/cs454-midterm-project$
```

Figure 1. Successful responses from `/health` and `/` inside the Ubuntu VM.

Docker Compose, Host Access, and Restart

From Windows PowerShell, I requested /health and / through 127.0.0.1:18081. Both endpoints returned HTTP 200.

```
PS C:\Users\sarah> curl.exe -i http://127.0.0.1:18081/health
HTTP/1.1 200 OK
Content-Type: application/json
Date: Wed, 21 Oct 2026 03:33:55 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"status":"ok"}
PS C:\Users\sarah> curl.exe -i http://127.0.0.1:18081/
HTTP/1.1 200 OK
Content-Type: application/json
Date: Wed, 21 Oct 2026 03:34:06 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"service":"cs554-midterm","message":"Hello from my midterm service!","hostn
ame":"51a355a22d67","requestCount":2}
PS C:\Users\sarah>
```

Figure 2. Successful requests from Windows to the application through NAT-forwarded port 18081.

Container Status and Logs

Inside the VM, `docker compose ps` showed the application container as healthy with port 8080 published. `docker compose logs --no-color --tail=10 app` showed `GET /health` requests with status 200.

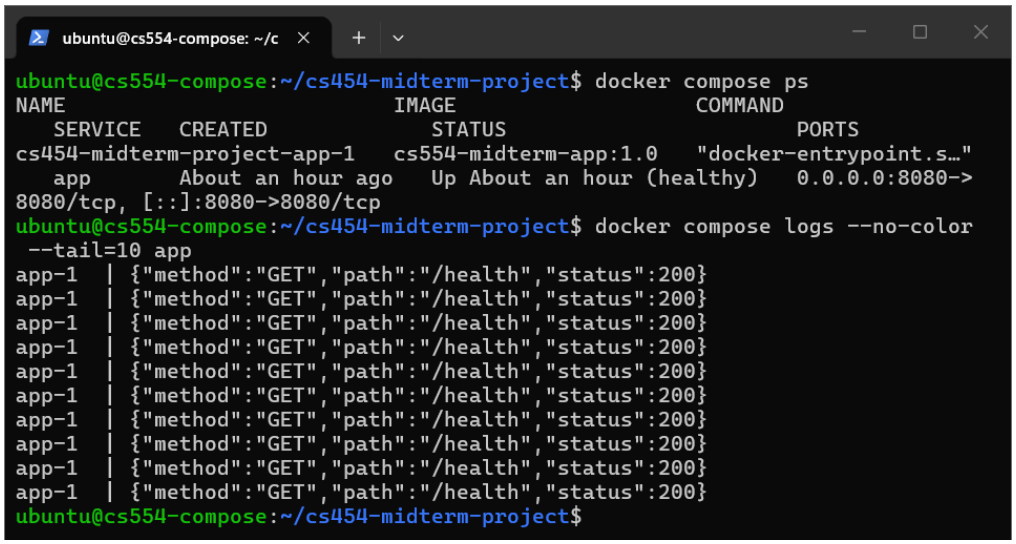


Figure 4. Docker Compose status and application logs showing successful health requests.

Counter Restart

Two consecutive requests to `/` returned counts of 3 and 4. After `docker compose restart app` and `docker compose up -d --wait`, the next request returned a count of 1. The container hostname remained unchanged.

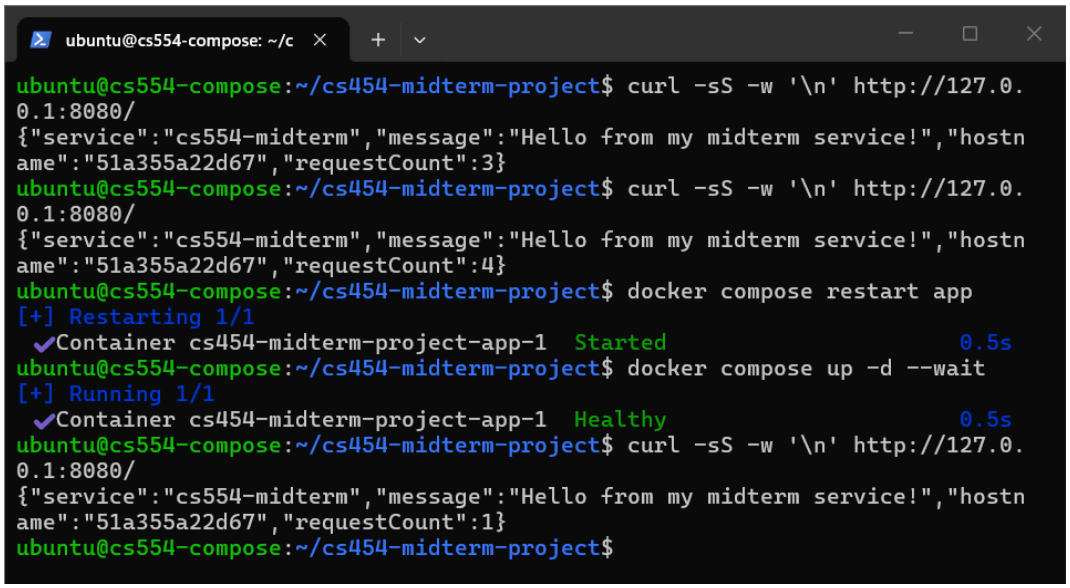


Figure 5. Request counter before and after restarting the Compose service.

Kubernetes Deployment and Pod Replacement

In Ubuntu WSL, `kubectl get deployment,pods,service -o wide` showed the Deployment with 2/2 Ready replicas. Both application Pods were Running and 1/1 Ready. The ClusterIP Service exposed port 80 and selected Pods labeled `app=cs554-midterm`.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get deployment,pods,service -o wide
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES                  SELECTOR
deployment.apps/cs554-midterm   2/2     2            2           11h   api          cs554-midterm-app:1.0   app=cs554-midterm

NAME                                    READY   STATUS    RESTARTS   AGE   IP           NODE                        NOMINATED NODE   READINESS GATES
pod/cs554-midterm-5f85d994bf-c86nn   1/1     Running   0          86m   10.42.0.34   k3d-cs554-midterm-server-0   <none>           <none>
pod/cs554-midterm-5f85d994bf-vnc72   1/1     Running   0          86m   10.42.0.36   k3d-cs554-midterm-server-0   <none>           <none>

NAME                  TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE   SELECTOR
service/cs554-midterm   ClusterIP   10.43.18.185   <none>        80/TCP    11h   app=cs554-midterm
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 6. Kubernetes Deployment with two Ready application replicas and a ClusterIP Service.

Requests through the Service

Using `kubectl exec`, I ran `wget` from midterm-client against `/health` and `/` through the Service name `cs554-midterm`. Both returned HTTP 200. The root response identified an application Pod by hostname.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec midterm-client -- wget -S -O - http://cs554-midterm/health
--2026-10-21 04:28:39--  http://cs554-midterm/health
Resolving cs554-midterm (cs554-midterm)... 10.43.18.185
Connecting to cs554-midterm (cs554-midterm)|10.43.18.185|:80... connected.
HTTP request sent, awaiting response... {"status":"ok"}
  HTTP/1.1 200 OK
  Content-Type: application/json
  Date: Wed, 21 Oct 2026 04:28:39 GMT
  Connection: keep-alive
  Keep-Alive: timeout=5
  Transfer-Encoding: chunked
Length: unspecified [application/json]
Saving to: 'STDOUT'

     0K                                                       1.90M=0s

2026-10-21 04:28:39 (1.90 MB/s) - written to stdout [15]

root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec midterm-client -- wget -S -O - http://cs554-midterm/
--2026-10-21 04:28:47--  http://cs554-midterm/
Resolving cs554-midterm (cs554-midterm)... 10.43.18.185
Connecting to cs554-midterm (cs554-midterm)|10.43.18.185|:80... connected.
HTTP request sent, awaiting response...
  HTTP/1.1 200 OK
  Content-Type: application/json
  Date: Wed, 21 Oct 2026 04:28:47 GMT
  Connection: keep-alive
  Keep-Alive: timeout=5
  Transfer-Encoding: chunked
Length: unspecified [application/json]
Saving to: 'STDOUT'

     0K                                                       20.2M=0s

2026-10-21 04:28:47 (20.2 MB/s) - written to stdout [131]

{"service":"cs554-midterm","message":"Hello from my midterm service!","hostname":"cs554-midterm-5f85d994bf-c86nn","requestCount":1}root@LAPTOP-B3C65M98:~/cs
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 7. Successful requests through the Kubernetes Service.

Pod Deletion and Replacement

Before deletion, both application Pods were Running and Ready.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get deployment cs554-midterm
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
cs554-midterm   2/2     2           2           11h
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pods -l app=cs554-midterm -o wide
NAME                             READY   STATUS    RESTARTS   AGE   IP           NODE                       NOMINATED NODE   READINESS GATES
cs554-midterm-5f85d994bf-c86nn   1/1     Running   0          96m   10.42.0.34   k3d-cs554-midterm-server-0   <none>           <none>
cs554-midterm-5f85d994bf-vnc72   1/1     Running   0          96m   10.42.0.36   k3d-cs554-midterm-server-0   <none>           <none>
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 8. Two Ready application Pods before deleting one replica.

I deleted `cs554-midterm-5f85d994bf-c86nn`. A new Pod, `cs554-midterm-5f85d994bf-qpxmv`, appeared, while `cs554-midterm-5f85d994bf-vnc72` remained. The Deployment returned to 2/2 Ready replicas.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get deployment cs554-midterm
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
cs554-midterm   2/2     2           2           11h
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pods -l app=cs554-midterm -o wide
NAME                             READY   STATUS    RESTARTS   AGE   IP           NODE                       NOMINATED NODE   READINESS GATES
cs554-midterm-5f85d994bf-c86nn   1/1     Running   0          96m   10.42.0.34   k3d-cs554-midterm-server-0   <none>           <none>
cs554-midterm-5f85d994bf-vnc72   1/1     Running   0          96m   10.42.0.36   k3d-cs554-midterm-server-0   <none>           <none>
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm delete pod cs554-midterm-5f85d994bf-c86nn
pod "cs554-midterm-5f85d994bf-c86nn" deleted from cs554-midterm namespace
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm rollout status deployment/cs554-midterm --timeout=120s
deployment "cs554-midterm" successfully rolled out
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get deployment cs554-midterm
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
cs554-midterm   2/2     2           2           11h
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pods -l app=cs554-midterm -o wide
NAME                             READY   STATUS    RESTARTS   AGE   IP           NODE                       NOMINATED NODE   READINESS GATES
cs554-midterm-5f85d994bf-qpxmv   1/1     Running   0          21s   10.42.0.38   k3d-cs554-midterm-server-0   <none>           <none>
cs554-midterm-5f85d994bf-vnc72   1/1     Running   0          98m   10.42.0.36   k3d-cs554-midterm-server-0   <none>           <none>
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 9. Kubernetes replaces a deleted application Pod and restores two Ready replicas.

CS 554 Resource and Readiness Configuration

`kubectl describe deployment cs554-midterm` showed CPU request `100m`, memory request `64Mi`, CPU limit `500m`, and memory limit `128Mi`. It also showed an exec readiness probe using `wget` against `/health`.

```
Pod Template:
  Labels:  app=cs554-midterm
  Containers:
   api:
    Image:      cs554-midterm-app:1.0
    Port:       8080/TCP (http)
    Host Port:  0/TCP (http)
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:      100m
      memory:   64Mi
    Readiness:  exec [wget -q -O /dev/null http://127.0.0.1:8080/health] delay=2s timeout=1s period=2s #success=1 #failure=2
    Environment:
      PORT:           8080
      SERVICE_NAME:   cs554-midterm
      MESSAGE:        Hello from my midterm service!
      NOT_READY_FILE:  /tmp/midterm-not-ready
    Mounts:           <none>
  Volumes:            <none>
  Node-Selectors:     <none>
  Tolerations:        <none>
```

Figure 10. Application container resource requests, limits, and readiness configuration.

Service Selector Failure and Readiness Testing

Service Baseline

Both application Pods were Ready and labeled `app=cs554-midterm`. The Service selector matched those labels, and its EndpointSlice listed both endpoints as ready. A request from `midterm-client` returned HTTP 200.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pods -l app=cs554-midterm --show-labels
NAME                          READY   STATUS    RESTARTS   AGE    LABELS
cs554-midterm-5f85d994bf-qpxmv   1/1     Running   0          19m    app=cs554-midterm,pod-template-hash=5f85d994bf
cs554-midterm-5f85d994bf-vnc72   1/1     Running   0          117m   app=cs554-midterm,pod-template-hash=5f85d994bf
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get service cs554-midterm -o wide
NAME           TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE   SELECTOR
cs554-midterm   ClusterIP   10.43.18.185   <none>        80/TCP    11h   app=cs554-midterm
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get endpointslices -l kubernetes.io/service-name=cs554-midterm -o yaml
apiVersion: v1
items:
- addressType: IPv4
  apiVersion: discovery.k8s.io/v1
  endpoints:
  - addresses:
    - 10.42.0.36
    conditions:
      ready: true
      serving: true
      terminating: false
    nodeName: k3d-cs554-midterm-server-0
    targetRef:
      kind: Pod
      name: cs554-midterm-5f85d994bf-vnc72
      namespace: cs554-midterm
      uid: 83706dc2-f5f7-4ded-a664-e3b598dd830d
  - addresses:
    - 10.42.0.38
    conditions:
      ready: true
      serving: true
      terminating: false
    nodeName: k3d-cs554-midterm-server-0
    targetRef:
      kind: Pod
      name: cs554-midterm-5f85d994bf-qpxmv
      namespace: cs554-midterm
      uid: ddb2998f-d761-4363-8cd5-25e3ab231c9e
  kind: EndpointSlice
  metadata:
    annotations:
      endpoints.kubernetes.io/last-change-trigger-time: "2026-10-21T04:33:24Z"
```

Figure 11a. Matching Pod labels and Service selector, with two ready endpoints.

```
      kubernetes.io/service-name: cs554-midterm
    name: cs554-midterm-lx8mq
    namespace: cs554-midterm
    ownerReferences:
    - apiVersion: v1
      blockOwnerDeletion: true
      controller: true
      kind: Service
      name: cs554-midterm
      uid: 37566bdf-b083-45b2-b440-f42a5c5417aa
    resourceVersion: "9689"
    uid: ca17d845-e2e9-45fc-8028-e960bff59597
  ports:
  - name: http
    port: 8080
    protocol: TCP
kind: List
metadata:
  resourceVersion: ""
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec midterm-client -- wget -S -T 3 -t 1 -O - http://cs554-midterm/
--2026-10-21 04:53:08--  http://cs554-midterm/
Resolving cs554-midterm (cs554-midterm)... 10.43.18.185
Connecting to cs554-midterm (cs554-midterm)|10.43.18.185|:80... connected.
HTTP request sent, awaiting response...
  HTTP/1.1 200 OK
  Content-Type: application/json
  Date: Wed, 21 Oct 2026 04:53:08 GMT
  Connection: keep-alive
  Keep-Alive: timeout=5
  Transfer-Encoding: chunked
Length: unspecified [application/json]
Saving to: 'STDOUT'

     0K                                                       {"service":"cs554-midterm","message":"Hello from my midterm service!","hostname":"cs554-midter
m-5f85d994bf-vnc72","requestCount":1} 19.8M=0s

2026-10-21 04:53:08 (19.8 MB/s) - written to stdout [131]

root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 11b. Successful Service request before introducing the selector mismatch.

Selector Failure

Using `kubectl patch service`, I changed the selector to `app=cs554-no-match`. Both Pods remained Running and Ready, but the Service had no matching endpoints. The client request failed with “Connection refused” and exit code 4.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm patch service cs554-midterm --type=merge -p '{"spec":{"selector":{"app":"cs554-no-mat
ch"}}}'
service/cs554-midterm patched
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pods -l app=cs554-midterm --show-labels
NAME                             READY   STATUS    RESTARTS   AGE    LABELS
cs554-midterm-5f85d994bf-qpxmv   1/1     Running   0          26m    app=cs554-midterm,pod-template-hash=5f85d994bf
cs554-midterm-5f85d994bf-vnc72   1/1     Running   0          124m   app=cs554-midterm,pod-template-hash=5f85d994bf
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get service cs554-midterm -o wide
NAME           TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE   SELECTOR
cs554-midterm   ClusterIP   10.43.18.185   <none>        80/TCP    11h   app=cs554-no-match
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get endpointslices -l kubernetes.io/service-name=cs554-midterm
NAME                 ADDRESSTYPE   PORTS     ENDPOINTS   AGE
cs554-midterm-lx8mq   IPv4         <unset>   <unset>     11h
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec midterm-client -- wget -S -T 3 -t 1 -O - http://cs554-midterm/
--2026-10-21 05:00:00--  http://cs554-midterm/
Resolving cs554-midterm (cs554-midterm)... 10.43.18.185
Connecting to cs554-midterm (cs554-midterm)|10.43.18.185|:80... failed: Connection refused.
command terminated with exit code 4
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 12. Selector mismatch prevents Service access while application Pods remain healthy.

Selector Recovery

Running `kubectl apply -f kubernetes/service.yaml` restored the selector to `app=cs554-midterm`. Both endpoints reappeared, and the client request returned HTTP 200.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm apply -f kubernetes/service.yaml
service/cs554-midterm configured
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get service cs554-midterm -o wide
NAME           TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE   SELECTOR
cs554-midterm   ClusterIP   10.43.18.185   <none>        80/TCP    11h   app=cs554-midterm
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get endpointslices -l kubernetes.io/service-name=cs554-midterm
NAME                 ADDRESSTYPE   PORTS   ENDPOINTS               AGE
cs554-midterm-lx8mq   IPv4         8080    10.42.0.36,10.42.0.38   11h
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec midterm-client -- wget -S -T 3 -t 1 -O - http://cs554-midterm/
--2026-10-21 05:02:54--  http://cs554-midterm/
Resolving cs554-midterm (cs554-midterm)... 10.43.18.185
Connecting to cs554-midterm (cs554-midterm)|10.43.18.185|:80... connected.
HTTP request sent, awaiting response...
  HTTP/1.1 200 OK
  Content-Type: application/json
  Date: Wed, 21 Oct 2026 05:02:54 GMT
  Connection: keep-alive
  Keep-Alive: timeout=5
  Transfer-Encoding: chunked
Length: unspecified [application/json]
Saving to: 'STDOUT'

     0K                                                       19.4M=0s

2026-10-21 05:02:54 (19.4 MB/s) - written to stdout [131]

{"service":"cs554-midterm","message":"Hello from my midterm service!","hostname":"cs554-midterm-5f85d994bf-qpxmv","requestCount":1}root@LAPTOP-B3C65M98:~/cs
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 13. Restoring the matching Service selector restores endpoints and HTTP access.

CS 554: Readiness Baseline

Both application Pods were Ready, and both endpoints showed `ready=true`. I recorded the selected Pod's UID, container ID, and restart count.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# POD=$(kubectl -n cs554-midterm get pods -l app=cs554-midterm -o jsonpath='{.items[0].metadata.name}')
root@LAPTOP-B3C65M98:~/cs454-midterm-project# echo "$POD"
cs554-midterm-5f85d994bf-qpxmv
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pods -l app=cs554-midterm -o wide
NAME                              READY   STATUS    RESTARTS   AGE    IP           NODE                          NOMINATED NODE   READINESS GATES
cs554-midterm-5f85d994bf-qpxmv    1/1     Running   0          33m    10.42.0.38   k3d-cs554-midterm-server-0   <none>           <none>
cs554-midterm-5f85d994bf-vnc72    1/1     Running   0          131m   10.42.0.36   k3d-cs554-midterm-server-0   <none>           <none>
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pod "$POD" -o jsonpath='UID: {.metadata.uid}{"\n"}Container: {.status.containerSt
atuses[0].containerID}{"\n"}Restarts: {.status.containerStatuses[0].restartCount}{"\n"}'
UID: ddb2998f-d761-4363-8cd5-25e3ab231c9e
Container: containerd://69c0b5662235eb5762c50ddf7fc1e8590a7e6c994717c45d0e1f2870a44fcab5
Restarts: 0
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get endpointslices -l kubernetes.io/service-name=cs554-midterm -o jsonpath='{range .i
tems[*].endpoints[*]}{.targetRef.name}{" ready="}{.conditions.ready}{"\n"}{end}'
cs554-midterm-5f85d994bf-vnc72 ready=true
cs554-midterm-5f85d994bf-qpxmv ready=true
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 14. Readiness baseline with two ready endpoints and the selected Pod's container identity.

One Replica NotReady

I created `/tmp/midterm-not-ready inside Pod qpxmv`. The Pod remained Running but became 0/1 Ready. Its endpoint showed `ready=false`, while `vnc72` remained `ready=true`. The selected Pod's UID and container ID were unchanged, and its restart count remained zero.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec "$POD" -- touch /tmp/midterm-not-ready
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm wait --for=condition=Ready=false "pod/$POD" --timeout=60s
pod/cs554-midterm-5f85d994bf-qpxmv condition met
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pods -l app=cs554-midterm -o wide
NAME                              READY   STATUS    RESTARTS   AGE    IP           NODE                          NOMINATED NODE   READINESS GATES
cs554-midterm-5f85d994bf-qpxmv    0/1     Running   0          37m    10.42.0.38   k3d-cs554-midterm-server-0   <none>           <none>
cs554-midterm-5f85d994bf-vnc72    1/1     Running   0          135m   10.42.0.36   k3d-cs554-midterm-server-0   <none>           <none>
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pod "$POD" -o jsonpath='UID: {.metadata.uid}{"\n"}Container: {.status.containerSt
atuses[0].containerID}{"\n"}Restarts: {.status.containerStatuses[0].restartCount}{"\n"}'
UID: ddb2998f-d761-4363-8cd5-25e3ab231c9e
Container: containerd://69c0b5662235eb5762c50ddf7fc1e8590a7e6c994717c45d0e1f2870a44fcab5
Restarts: 0
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
kubectl -n cs554-midterm get endpointslices -l kubernetes.io/service-name=cs554-midterm -o jsonpath='{range .items[*].endpoints[*]}{.targetRef.name}{" ready
="}{.conditions.ready}{"\n"}{end}'
cs554-midterm-5f85d994bf-vnc72 ready=true
cs554-midterm-5f85d994bf-qpxmv ready=false
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 15a. One replica becomes NotReady while remaining Running.

.

HTTP Responses During The Readiness Failure

A direct request to `/health` on `qpxmv` returned HTTP 503. A direct request to `/` on the same Pod returned HTTP 200. A request through the Service returned HTTP 200 from `vnc72`.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec "$POD" -- wget -q -S -T 3 -t 1 -O - http://127.0.0.1:8080/health
  HTTP/1.1 503 Service Unavailable
  Content-Type: application/json
  Date: Wed, 21 Oct 2026 05:12:35 GMT
  Connection: keep-alive
  Keep-Alive: timeout=5
  Transfer-Encoding: chunked
command terminated with exit code 8
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec "$POD" -- wget -q -S -T 3 -t 1 -O - http://127.0.0.1:8080/
{"service":"cs554-midterm","message":"Hello from my midterm service!","hostname":"cs554-midterm-5f85d994bf-qpxmv","requestCount":2}  HTTP/1.1 200 OK
  Content-Type: application/json
  Date: Wed, 21 Oct 2026 05:12:41 GMT
  Connection: keep-alive
  Keep-Alive: timeout=5
  Transfer-Encoding: chunked
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec midterm-client -- wget -q -S -T 3 -t 1 -O - http://cs554-midterm/
  HTTP/1.1 200 OK
  Content-Type: application/json
  Date: Wed, 21 Oct 2026 05:12:46 GMT
  Connection: keep-alive
  Keep-Alive: timeout=5
  Transfer-Encoding: chunked
{"service":"cs554-midterm","message":"Hello from my midterm service!","hostname":"cs554-midterm-5f85d994bf-vnc72","requestCount":2}root@LAPTOP-B3C65M98:~/cs
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 15b. The NotReady Pod continues serving root requests while the Service reaches the other ready replica.

Readiness Recovery

I removed the marker file. Both Pods returned to Ready, and both endpoints showed `ready=true`. A Service request returned HTTP 200 from `qpxmv`. Its UID and container ID remained unchanged, with zero restarts.

```
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec "$POD" -- rm -f /tmp/midterm-not-ready
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm wait --for=condition=Ready "pod/$POD" --timeout=60s
pod/cs554-midterm-5f85d994bf-qpxmv condition met
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pods -l app=cs554-midterm -o wide
NAME                              READY   STATUS    RESTARTS   AGE    IP           NODE                        NOMINATED NODE   READINESS GATES
cs554-midterm-5f85d994bf-qpxmv   1/1     Running   0          42m    10.42.0.38   k3d-cs554-midterm-server-0   <none>           <none>
cs554-midterm-5f85d994bf-vnc72   1/1     Running   0          140m   10.42.0.36   k3d-cs554-midterm-server-0   <none>           <none>
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get pod "$POD" -o jsonpath='UID: {.metadata.uid}{"\n"}Container: {.status.containerSt
atuses[0].containerID}{"\n"}Restarts: {.status.containerStatuses[0].restartCount}{"\n"}'
UID: ddb2998f-d761-4363-8cd5-25e3ab231c9e
Container: containerd://69c0b5662235eb5762c50ddf7fc1e8590a7e6c994717c45d0e1f2870a44fcab5
Restarts: 0
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm get endpointslices -l kubernetes.io/service-name=cs554-midterm -o jsonpath='{range .i
tems[*].endpoints[*]}{.targetRef.name}{" ready="}{.conditions.ready}{"\n"}{end}'
cs554-midterm-5f85d994bf-vnc72 ready=true
cs554-midterm-5f85d994bf-qpxmv ready=true
root@LAPTOP-B3C65M98:~/cs454-midterm-project# kubectl -n cs554-midterm exec midterm-client -- wget -q -S -T 3 -t 1 -O - http://cs554-midterm/
{"service":"cs554-midterm","message":"Hello from my midterm service!","hostname":"cs554-midterm-5f85d994bf-qpxmv","requestCount":3}  HTTP/1.1 200 OK
  Content-Type: application/json
  Date: Wed, 21 Oct 2026 05:15:57 GMT
  Connection: keep-alive
  Keep-Alive: timeout=5
  Transfer-Encoding: chunked
root@LAPTOP-B3C65M98:~/cs454-midterm-project#
```

Figure 16. Removing the readiness marker restores both ready endpoints without restarting the selected container.